KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

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Enabling agents to autonomously regulate knowledge utilization through situational self-awareness

Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection Hurts Efficiency

Direct Trajectory Imitation

ReAct

Blindly injects gold trajectories regardless of whether the agent already knows the correct action. Leads to over-reliance on memorized patterns.

Trial-and-Error Refinement

Reflexion

Indiscriminately applies external feedback at every step, even when the agent could self-correct without it. Wastes inference budget.

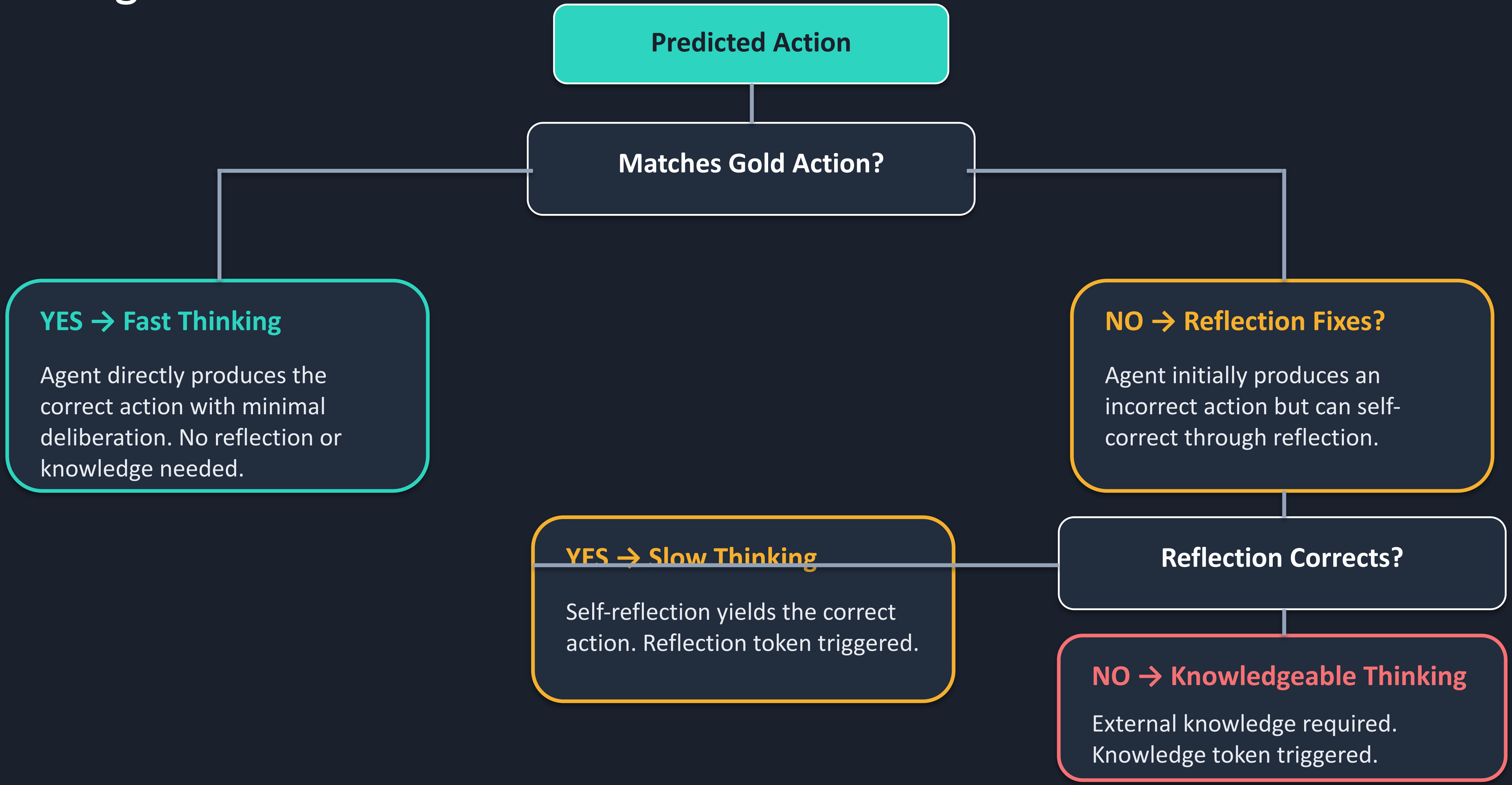
Knowledge-Augmented Planning

KnowAgent / WKM

Flood irrigation with domain knowledge — 100% knowledge injection even when the agent is already capable. Fragile and costly.

Humans possess situational self-awareness — the metacognitive ability to assess whether they can handle a situation directly, need to pause and reflect, or must seek external help. KnowSelf brings this capability to LLM agents.

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking



KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

Step 1

Self-Awareness Data Construction

- Agent self-explores environments
- Heuristic criterion classifies each step:
  - Match gold action → Fast thinking
  - Reflection fixes → Slow thinking
  - Otherwise → Knowledgeable thinking
- Special tokens inserted to mark situation type

Step 2

Two-Stage Training

- Stage A: Supervised Fine-Tuning (SFT)
  - Teaches initial self-awareness patterns
  - Learns to generate situation tokens
- Stage B: Reward-based Policy Optimization (RPO)
  - Refines token generation accuracy
  - Maximizes reward for correct mode selection

Step 3

Self-Aware Inference

- At each decision step, agent generates a special token:
  - [FAST] → Direct action output
  - [SLOW] → Trigger reflection loop
  - [KNOW] → Retrieve from external KB
- Selective activation minimizes knowledge queries
- Dynamic cognitive resource regulation

The entire pipeline requires only heuristic labeling on self-explored trajectories — no expensive human annotation or massive external feedback.

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Task: Put a hot mug in cabinet

OpenAI-O1 Failure

Incorrectly tries to take an egg from the microwave after opening it. Fails to recognize the correct object to heat.

KnowSelf (Llama-8B) Success

Emits a [SLOW] Reflection token to realize it should heat the mug using the open microwave. Self-corrects without external knowledge.

Task: Put a saltshaker in drawer

DeepSeek-R1 Failure

Reflects that the drawer is open and tries to close it first — which fails. Gets stuck on an irrelevant sub-goal.

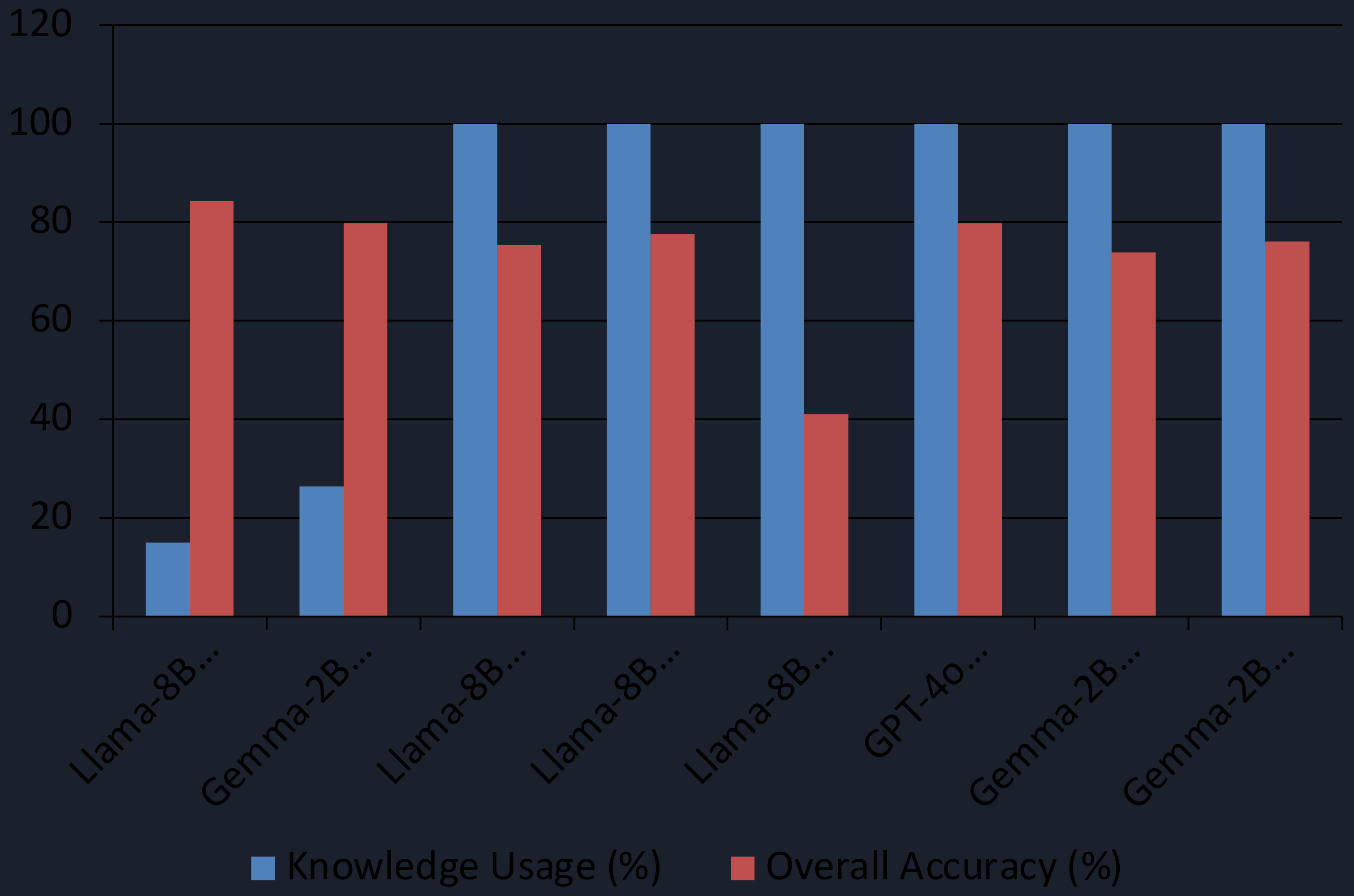
KnowSelf (Llama-8B) Success

Applies a [KNOW] Knowledge token noting the agent should place the object without removing unrelated items. Correctly uses external knowledge.

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	**KnowSelf**	**15.01%**	**91.67**	**87.10**	**91.30**	**85.71**	**77.78**	**64.71**	**84.33**
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	**KnowSelf**	**26.41%**	**87.50**	**93.55**	**73.91**	**76.19**	**83.33**	**52.94**	**79.85**

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage Over Full-Knowledge Methods



15% Knowledge → 84.33% Accuracy

Llama-8B KnowSelf uses only 15% external knowledge yet surpasses GPT-4o ExpeL (79.85%) which uses 100%.

26% Knowledge → 79.85% Accuracy

Gemma-2B KnowSelf matches GPT-4o ExpeL with just 26% knowledge, proving scalability to smaller models.

100% Knowledge Often Hurts

ExpeL on Llama-8B drops to 41.04% despite full knowledge injection — indiscriminate augmentation degrades performance.

Selective > Blanket

KnowSelf's situational self-awareness provides principled cognitive resource regulation, analogous to human metacognition.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

1. Indiscriminate knowledge injection is wasteful and suboptimal

Current methods (ReAct, Reflexion, KnowAgent, WKM) flood agents with trajectories, feedback, or knowledge regardless of situational need — causing fragile planning and inflated costs.

2. Agents can learn situational self-awareness via lightweight special tokens

KnowSelf uses heuristic labeling on self-explored trajectories to teach agents when to act fast, reflect slowly, or seek knowledge — no expensive human annotation required.

3. KnowSelf with Llama-8B outperforms strong baselines including GPT-4o

84.33% overall ALFWorld accuracy surpasses GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%), demonstrating that selective knowledge beats blanket augmentation.

4. Only 15–26% knowledge usage needed versus 100% for competing methods

KnowSelf queries external knowledge on a fraction of steps while maintaining or exceeding the accuracy of full-knowledge methods — a dramatic efficiency gain.

5. The paradigm generalizes across model scales

Even Gemma-2B reaches 79.85% with KnowSelf, matching GPT-4o ExpeL and exceeding all other Gemma-2B baselines. Self-awareness scales down effectively.

Future Directions: Extend to multi-agent collaboration, continuous online self-awareness updates, and broader embodied AI benchmarks beyond ALFWorld.